Shared Protocol

The common layer underneath every board. The overview, single-stick, dual-stick, and robot docs each layer their own behavior on top of what's defined here.

TRANSPORT

- Fixed WiFi channel (channel 1) for every device, so peers can hear each other without joining a network.
- Peers are registered explicitly via `esp_now_add_peer()` using each other's hardcoded MAC address — clean unicast, no broadcast noise.
- No pairing handshake, no IP stack, no dropped sessions to manage. Any bare ESP32 flashed with the struct definitions and the target's MAC address can join the system — no authentication, no proprietary handshake.

PACKET TYPING — BY LENGTH

There is no header byte. Every packet is a small, fixed-size, packed C struct, and receivers tell packet types apart purely by their length on the wire — zero-overhead message typing.

```
typedef struct __attribute__((packed)) {
  uint16_t x;     // joystick X, 0-4095
  uint16_t y;     // joystick Y, 0-4095
  uint32_t seq;   // rolling sequence #
  uint32_t ms;    // sender timestamp
} JoyPacket;      // 12 bytes — drive

typedef struct __attribute__((packed)) {
  uint8_t  state; // 0 = off, 1 = on
  uint32_t seq;
} LightPacket;    // 5 bytes — accessory event
```

STRUCT	SIZE	CARRIES	CADENCE
`JoyPacket`	12 bytes	Drive (X/Y)	Continuous, ~5ms
`LightPacket`	5 bytes	One-shot accessory event	Fired on trigger, resent in a burst

TALKING BACK: THE HEARTBEAT

The motor receiver sends a periodic heartbeat back to any listening controller, roughly every 500ms, carrying RSSI, relay state, and uptime. A screen-equipped controller (dual-stick) renders this as a live HUD; a simple controller (single-stick) just never asks for it. The robot doesn't need to know or care which is connected — it always sends the heartbeat, and only a listening controller does anything with it.

RELIABILITY, HANDLED AT THE APPLICATION LAYER

There's no ACK/retry layer in ESP-NOW itself, so reliability is designed per packet type:

- **Continuous streams (drive)** are self-healing — a dropped packet is superseded ~5ms later by the next one. A receiver-side timeout is the fail-safe stop.
- **One-shot events (light toggles)** are made reliable by resending a few times in a burst rather than waiting on an ACK.

DEVICE ROLES & MAC ADDRESSES

DEVICE	ROLE	MAC ADDRESS
Motor receiver (robot)	Drives motors, relay/lights	`C0:CD:D6:CA:0E:4C`
Relay receiver (robot)	Switches lights/accessories	`D4:E9:F4:E6:A7:B8`
Single-joystick controller	Sender — drive + lights	—
Dual-joystick controller	Sender + receiver — telemetry	`D4:E9:F4:E6:07:F8`

WHAT EACH BOARD ADDS ON TOP OF THIS

- **Single-stick controller** — one-way, fire-and-forget. Sends `JoyPacket` continuously and `LightPacket` on double-click. Never listens for the heartbeat.
- **Dual-stick controller** — bidirectional. Adds a second stick for lights/turn signals and listens for the heartbeat to drive a TFT HUD.
- **Robot (motor + relay receivers)** — the only side that transmits the heartbeat. One fixed firmware image regardless of which controller is talking to it.